# 2.63 Numerical Methods: Mid-ordinate Rule

Worked Solutions

**1)**

The shaded rectangle has width 0.5 and height $f(1.75) = \tan(0.4 \times 1.75)$.

$$0.5 \times \tan(0.4 \times 1.75) = 0.421 \text{ units}^2 \quad (3 \text{ SF}).$$

**2 a)**

For $h = 2$, the midpoint is $x_{1-\frac{1}{2}} = 1$.

$$\int_0^2 e^{-x^2}\,\mathrm{d}x \approx 2e^{-1^2} = 0.735759 \quad (6 \text{ DP}).$$

**2 b)**

For $h = 1$, the midpoints are 0.5 and 1.5.

$$\int_0^2 e^{-x^2}\,\mathrm{d}x \approx 1\left(e^{-0.5^2} + e^{-1.5^2}\right) = 0.884200 \quad (6 \text{ DP}).$$

**2 c)**

For $h = 0.5$, the midpoints are 0.25, 0.75, 1.25 and 1.75.

$$\begin{aligned}\int_0^2 e^{-x^2}\,\mathrm{d}x &\approx 0.5\left(e^{-0.25^2} + e^{-0.75^2} + e^{-1.25^2} + e^{-1.75^2}\right)\\ &= 0.882789 \quad (6 \text{ DP}).\end{aligned}$$

**3 a)**

With one strip, the midpoint is 1.5 and the width is 1.

$$\int_1^2 \frac{1}{1-\ln x}\,\mathrm{d}x \approx 1\left(\frac{1}{1-\ln 1.5}\right) = 1.681987 \quad (6 \text{ DP}).$$

**3 b)**

With two strips, the midpoints are 1.25 and 1.75 and the width is 0.5.

$$\begin{aligned}\int_1^2 \frac{1}{1-\ln x}\,\mathrm{d}x &\approx 0.5\left(\frac{1}{1-\ln 1.25} + \frac{1}{1-\ln 1.75}\right)\\ &= 1.778992 \quad (6 \text{ DP}).\end{aligned}$$

**3 c)**

With four strips, the midpoints are 1.125, 1.375, 1.625 and 1.875 and the width is 0.25.

$$\begin{aligned}\int_1^2 \frac{1}{1-\ln x}\,\mathrm{d}x &\approx 0.25\left(\frac{1}{1-\ln 1.125} + \frac{1}{1-\ln 1.375} + \frac{1}{1-\ln 1.625} + \frac{1}{1-\ln 1.875}\right)\\ &= 1.809250 \quad (6 \text{ DP}).\end{aligned}$$

**4 a)**

The strip width is $h = 0.5$, and the midpoints are 0.25, 0.75, 1.25 and 1.75.

$$\begin{aligned}\int_0^2 \frac{\sin x}{x}\,\mathrm{d}x &\approx 0.5\left(\frac{\sin 0.25}{0.25} + \frac{\sin 0.75}{0.75} + \frac{\sin 1.25}{1.25} + \frac{\sin 1.75}{1.75}\right)\\ &= 1.61 \quad (3\text{ SF}).\end{aligned}$$

**4 b)**

Increase the number of strips used.

**5)**

The five strips each have width 2. The midpoints are 1, 3, 5, 7 and 9, and the corresponding velocities are 12.8, 17.2, 20, 21.2 and 20.8.

$$\int_0^{10} (10 + 3t - 0.2t^2)\,\mathrm{d}t \approx 2(12.8 + 17.2 + 20 + 21.2 + 20.8) = 184 \text{ metres.}$$

**6 a)**

With three rectangles, the strip width is 1 and the midpoints are 1.5, 2.5 and 3.5.

$$\int_1^4 \ln x\,\mathrm{d}x \approx 1(\ln 1.5 + \ln 2.5 + \ln 3.5) = 2.5745 \quad (4\text{ DP}).$$

**6 b)**

Let $u = \ln x$ and $\mathrm{d}v/\mathrm{d}x = 1$, so $\mathrm{d}u/\mathrm{d}x = 1/x$ and $v = x$.

$$\begin{aligned}\int_1^4 \ln x\,\mathrm{d}x &= [x\ln x]_1^4 - \int_1^4 \frac{x}{x}\,\mathrm{d}x\\ &= [x\ln x - x]_1^4 = (4\ln 4 - 4) - (\ln 1 - 1)\\ &= -3 + 4\ln 4.\end{aligned}$$

**6 c)**

$$\frac{2.5745 - (-3 + 4\ln 4)}{-3 + 4\ln 4} = 0.0115.$$

**7 a)**

$$\int_a^b f(x)\,\mathrm{d}x \approx (b-a) f\left(\frac{a+b}{2}\right).$$

**7 b)**

The tangent line $PQ$ touches the curve at the midpoint $x = \dfrac{a+b}{2}$. Its gradient is $f'\left(\dfrac{a+b}{2}\right)$. Let $f'\left(\dfrac{a+b}{2}\right) = m$. Then the equation of the tangent is

$$y - f\left(\frac{a+b}{2}\right) = m\left(x - \frac{a+b}{2}\right).$$

At $x = a$,

$$y - f\left(\frac{a+b}{2}\right) = m\left(a - \frac{a+b}{2}\right) = m\left(\frac{a-b}{2}\right),$$

so $y = m\left(\frac{a-b}{2}\right) + f\left(\frac{a+b}{2}\right)$. At $x = b$,

$$y - f\left(\frac{a+b}{2}\right) = m\left(b - \frac{a+b}{2}\right) = m\left(\frac{b-a}{2}\right),$$

so $y = m\left(\frac{b-a}{2}\right) + f\left(\frac{a+b}{2}\right)$. The area under the tangent line is

$$\begin{aligned}
&\frac{1}{2}\left(m\left(\frac{a-b}{2}\right) + f\left(\frac{a+b}{2}\right) + m\left(\frac{b-a}{2}\right) + f\left(\frac{a+b}{2}\right)\right)(b-a) \\
&\quad = \frac{1}{2}\left(2f\left(\frac{a+b}{2}\right)\right)(b-a) \\
&\quad = f\left(\frac{a+b}{2}\right)(b-a),
\end{aligned}$$

which is the same as the mid-ordinate rule with one strip. Since this trapezium is an under-estimate, the mid-ordinate rule estimate must be an under-estimate.

Alternatively, the tangent line $PQ$ touches the curve at the midpoint $x = \frac{a+b}{2}$. The height of the trapezium at the midpoint is $f\left(\frac{a+b}{2}\right)$. The tangent is linear and symmetric about the midpoint, so the mean of the endpoint heights equals the midpoint height. Therefore the area is $f\left(\frac{a+b}{2}\right)(b-a)$, which is the same as the mid-ordinate rule with one strip. Since this trapezium is an under-estimate, the mid-ordinate rule estimate must be an under-estimate.

**8)**

$$\int_a^b f(x)\,\mathrm{d}x \approx (b-a)f\left(\frac{a+b}{2}\right).$$

Suppose there is a tangent line which touches the curve at the midpoint $x = \frac{a+b}{2}$. The height of the trapezium at the midpoint is $f\left(\frac{a+b}{2}\right)$. The tangent is linear and symmetric about the midpoint, so the mean of the endpoint heights equals the midpoint height.

$$\therefore f\left(\frac{a+b}{2}\right)(b-a),$$

which is the same as the mid-ordinate rule with one strip. Since this trapezium is an over-estimate, the mid-ordinate rule estimate must be an over-estimate.

**9 a)**

The strip width is 0.5 and the midpoints are 0.25 and 0.75.

$$\int_0^1 3\cos x\,\mathrm{d}x \approx 0.5(3\cos 0.25 + 3\cos 0.75) = 2.55 \quad (3\text{ SF}).$$

**9 b)**

The curve is concave (concave downwards) for $0 \leq x \leq 1$, so the estimate is an over-estimate.

**10 a)**

The strip width is 0.25, and the midpoints are 1.125, 1.375, 1.625 and 1.875.

$$\begin{aligned}\int_1^2 \sinh x \, \mathrm{d}x &\approx 0.25(\sinh 1.125 + \sinh 1.375 + \sinh 1.625 + \sinh 1.875) \\ &= 2.21 \quad (3 \text{ SF}).\end{aligned}$$

**10 b)**

The curve is convex (concave upwards) for $1 \leq x \leq 2$, so the estimate is an under-estimate.

**11 a)**

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $\sqrt{x^2-1}$ | 0 | $\sqrt{3}$ | $\sqrt{8}$ | $\sqrt{15}$ |

$$\begin{aligned}\int_1^4 \sqrt{x^2-1} \, \mathrm{d}x &\approx \frac{1}{2} \times 1 \left(0 + \sqrt{15} + 2(\sqrt{3} + \sqrt{8})\right) \\ &= 6.4970 \quad (4 \text{ DP}).\end{aligned}$$

**11 b)**

| $x_{i-\frac{1}{2}}$ | 1.5 | 2.5 | 3.5 |
|---|---|---|---|
| $\sqrt{x_{i-\frac{1}{2}}^2 - 1}$ | $\frac{\sqrt{5}}{2}$ | $\frac{\sqrt{21}}{2}$ | $\frac{3\sqrt{5}}{2}$ |

$$\int_1^4 \sqrt{x^2-1} \, \mathrm{d}x \approx 1 \left(\frac{\sqrt{5}}{2} + \frac{\sqrt{21}}{2} + \frac{3\sqrt{5}}{2}\right) = 6.7634 \quad (4 \text{ DP}).$$

**11 c)**

Since the curve is concave (concave downwards), the trapezium rule under-estimates the area.

**11 d)**

Since the curve is concave (concave downwards), the mid-ordinate rule over-estimates the area.

**12 a)**

| $x$ | 2 | 4 | 6 | 8 | 10 |
|---|---|---|---|---|---|
| $1/\ln x$ | $1/\ln 2$ | $1/\ln 4$ | $1/\ln 6$ | $1/\ln 8$ | $1/\ln 10$ |

$$\begin{aligned}\int_2^{10} \frac{1}{\ln x} \, \mathrm{d}x &\approx \frac{1}{2} \times 2 \left(\frac{1}{\ln 2} + \frac{1}{\ln 10} + 2\left(\frac{1}{\ln 4} + \frac{1}{\ln 6} + \frac{1}{\ln 8}\right)\right) \\ &= 5.3977 \quad (4 \text{ DP}).\end{aligned}$$

**12 b)**

| $x_{i-\frac{1}{2}}$ | 3 | 5 | 7 | 9 |
|---|---|---|---|---|
| $1/\ln x_{i-\frac{1}{2}}$ | $1/\ln 3$ | $1/\ln 5$ | $1/\ln 7$ | $1/\ln 9$ |

$$\int_2^{10} \frac{1}{\ln x}\,\mathrm{d}x \approx 2\left(\frac{1}{\ln 3}+\frac{1}{\ln 5}+\frac{1}{\ln 7}+\frac{1}{\ln 9}\right) = 5.0012 \quad (4 \text{ DP}).$$

**12 c)**

$$f'(x) = \frac{0-1}{x(\ln x)^2} = -\frac{1}{x(\ln x)^2}.$$

$$f''(x) = \frac{(\ln x)^2 + 2\ln x}{x^2(\ln x)^4} = \frac{\ln x + 2}{x^2(\ln x)^3}.$$

Since $\ln x + 2 > 0$ for all $x > 0$ and $x^2(\ln x)^3 > 0$ for $x > 1$, in the interval $2 \le x \le 10$ we have $f''(x) > 0$. So $f'(x)$ is always increasing for $2 \le x \le 10$.

**12 d)**

Since the curve is convex (concave upwards), the trapezium rule over-estimates the area.

**12 e)**

Since the curve is convex (concave upwards), the mid-ordinate rule under-estimates the area.

**13 a)**

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $e^x/x$ | $e$ | $e^2/2$ | $e^3/3$ |

$$\int_1^3 \frac{e^x}{x}\,\mathrm{d}x \approx \frac{1}{2}\times 1\left(e+\frac{e^3}{3}+2\left(\frac{e^2}{2}\right)\right) = 8.40126 \quad (5 \text{ DP}).$$

**13 b)**

| $x_{i-\frac{1}{2}}$ | 1.5 | 2.5 |
|---|---|---|
| $e^{x_{i-\frac{1}{2}}}/x_{i-\frac{1}{2}}$ | $e^{1.5}/1.5$ | $e^{2.5}/2.5$ |

$$\int_1^3 \frac{e^x}{x}\,\mathrm{d}x \approx 1\left(\frac{e^{1.5}}{1.5}+\frac{e^{2.5}}{2.5}\right) = 7.86079 \quad (5 \text{ DP}).$$

**13 c)**

The trapezium rule under-estimates the area for $1 \le x \le 3$ and the mid-ordinate rule over-estimates the area for $1 \le x \le 3$, so the curve is convex (concave upwards) for $1 \le x \le 3$.

**14 a)**

$\pi \displaystyle\int_0^2 (\ln(x+1))^2\,\mathrm{d}x.$

**14 b)**

| $x$ | 0 | 0.5 | 1 | 1.5 | 2 |
|---|---|---|---|---|---|
| $x_{i-\frac{1}{2}}$ | | 0.25 | 0.75 | 1.25 | 1.75 |
| $(\ln(x_{i-\frac{1}{2}}+1))^2$ | | $(\ln 1.25)^2$ | $(\ln 1.75)^2$ | $(\ln 2.25)^2$ | $(\ln 2.75)^2$ |

$$V \approx 0.5\pi\left((\ln 1.25)^2 + (\ln 1.75)^2 + (\ln 2.25)^2 + (\ln 2.75)^2\right)$$
$$= 3.21 \text{ units}^3 \quad (3 \text{ SF}).$$

**15 a)**

| $x$ | 0 | 0.5 | 1 |
|---|---|---|---|
| $x_{i-\frac{1}{2}}$ | | 0.25 | 0.75 |
| $\sin(x^2_{i-\frac{1}{2}})$ | | $\sin(0.25^2)$ | $\sin(0.75^2)$ |

$$\int_0^1 \sin(x^2)\,\mathrm{d}x \approx 0.5(\sin(0.25^2) + \sin(0.75^2)) = 0.2979 \quad (4 \text{ DP}).$$

**15 b)**

$f(x) = 3\sin((x-2)^2)$.

**15 c)**

$$\int_2^3 \sin(x^2)\,\mathrm{d}x \approx 3 \times 0.2979 = 0.894 \quad (3 \text{ DP}).$$

**15 d)**

| $x$ | 2 | 2.5 | 3 |
|---|---|---|---|
| $x_{i-\frac{1}{2}}$ | | 2.25 | 2.75 |
| $3\sin((x_{i-\frac{1}{2}}-2)^2)$ | | $3\sin(0.0625)$ | $3\sin(0.5625)$ |

$$\int_0^2 \sin(x^2)\,\mathrm{d}x \approx 0.5(3\sin(0.0625) + 3\sin(0.5625)) = 0.894 \quad (3 \text{ DP}).$$

**16 a)**

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $x_{i-\frac{1}{2}}$ | | 0.5 | 1.5 | 2.5 |
| $\sqrt{81-3x^3_{i-\frac{1}{2}}}$ | | $\frac{\sqrt{1290}}{4}$ | $\frac{9\sqrt{14}}{4}$ | $\frac{\sqrt{546}}{4}$ |

$$\int_0^3 \sqrt{81-3x^3}\,\mathrm{d}x \approx 1\left(\frac{\sqrt{1290}}{4} + \frac{9\sqrt{14}}{4} + \frac{\sqrt{546}}{4}\right) = 23.24 \quad (2 \text{ DP}).$$

**16 b)**

$$f(x) = (81 - 3x^3)^{1/2}, \qquad f'(x) = -\frac{9}{2}x^2(81 - 3x^3)^{-1/2}.$$

The tangent line:

$$f'(2) = -\frac{6\sqrt{57}}{19}.$$
$$y - \sqrt{57} = -\frac{6\sqrt{57}}{19}(x - 2).$$
$$19y - 19\sqrt{57} = -6\sqrt{57}x + 12\sqrt{57}.$$
$$19y + 6\sqrt{57}x = 31\sqrt{57}.$$

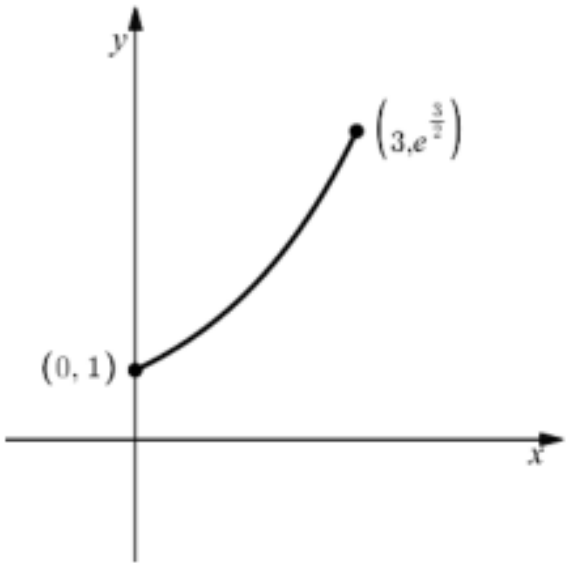
**16 c)**

When $x = 0$, $y = \dfrac{31\sqrt{57}}{19}$. When $y = 0$, $x = \dfrac{31}{6}$. So the area is

$$\frac{1}{2} \times \frac{31\sqrt{57}}{19} \times \frac{31}{6} - 23.24 = 8.58 \text{ units}^2 \quad (3 \text{ SF}).$$

**17 a)**

Sketch of $y = e^{x/2}$ for $0 \leq x \leq 3$.




**17 b)**

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $x_{i-\frac{1}{2}}$ | | 0.5 | 1.5 | 2.5 |
| $e^{x_{i-\frac{1}{2}}/2}$ | | $e^{1/4}$ | $e^{3/4}$ | $e^{5/4}$ |

$$\int_0^3 e^{x/2}\,\mathrm{d}x \approx 1(e^{1/4} + e^{3/4} + e^{5/4}) = 6.891368 \text{ units}^2 \quad (6 \text{ DP}).$$

**17 c)**

$$\frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x = \frac{1}{3}\int_0^3 e^{x/2}\,\mathrm{d}x \approx \frac{1}{3}\times 6.891368 = 2.29712 \quad (5\text{ DP}).$$

**18 a)**

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3}.$$

**18 b)**

| $x$ | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 |
|---|---|---|---|---|---|
| $x_{i-\frac{1}{2}}$ | | 0.15 | 0.25 | 0.35 | 0.45 |
| $x-\frac{x^2}{2}+\frac{x^3}{3}$ | | 0.139875 | 0.223958 | 0.303042 | 0.379125 |

$$\int_{0.1}^{0.5}\left(x-\frac{x^2}{2}+\frac{x^3}{3}\right)\mathrm{d}x \approx 0.1(0.139875+0.223958+0.303042+0.379125) = \frac{523}{5000}.$$

**19 a)**

$$xe^x = x\left(1+x+\frac{x^2}{2}\right) = x+x^2+\frac{x^3}{2}.$$

**19 b)**

| $x$ | 0 | 0.05 | 0.1 | 0.15 | 0.2 | 0.25 | 0.3 |
|---|---|---|---|---|---|---|---|
| $x_{i-\frac{1}{2}}$ | | 0.025 | 0.075 | 0.125 | 0.175 | 0.225 | 0.275 |
| $x+x^2+\frac{x^3}{2}$ | | 0.0256328 | 0.0808359 | 0.141602 | 0.208305 | 0.28132 | 0.361023 |

$$\int_0^{0.3}\left(x+x^2+\frac{x^3}{2}\right)\mathrm{d}x \approx 0.05(0.0256328+0.0808359+0.141602+0.208305+0.28132+0.361023) = 0.0549 \quad (4\text{ DP}).$$